Propeller Tachometer Generator – Maintenance Practices

1. Adjustment/Test

A. Bench Test Tachometer Generator

Equipment Required:

- Variable speed motor
- Drive speed indicator
- Tachometer generator output indicator (percent rpm)
- Mounting fixture and wiring
- Motor control switch
- HI–RPM–LOAD (Three 40 ohm, Y–connected resistances)
- LO–RPM–LOAD (Three 20 ohm, Y–connected resistances)
- Load ON–OFF switches (2)
- Multimeter

1) Test Continuity

a) Using multimeter (set to OHMS R x 1) measure continuity resistance of tachometer generator between each pair of stator windings A–B, A–C and B–C (C–case). Readings must be within 2.0 ohms of each other and each reading must be less than 50 ohms.

2) Test Rotation

a) Connect tachometer generator to output indicator (refer to Figure 1).

Note Tachometer generator is rotated in a CCW direction, viewed from the shaft end.

b) Operate motor and check output indicator pointer moves in a clockwise direction as motor speed is increased.

c) Shutdown motor.

3) Test Voltage Output

a) Disconnect wiring from output indicator (refer to Para 1.A.2).a)) and connect to test point terminals (refer to Figure 1).

b) Ensure LO–RPM–LOAD switch is OFF, and select HI–RPM–LOAD switch ON.

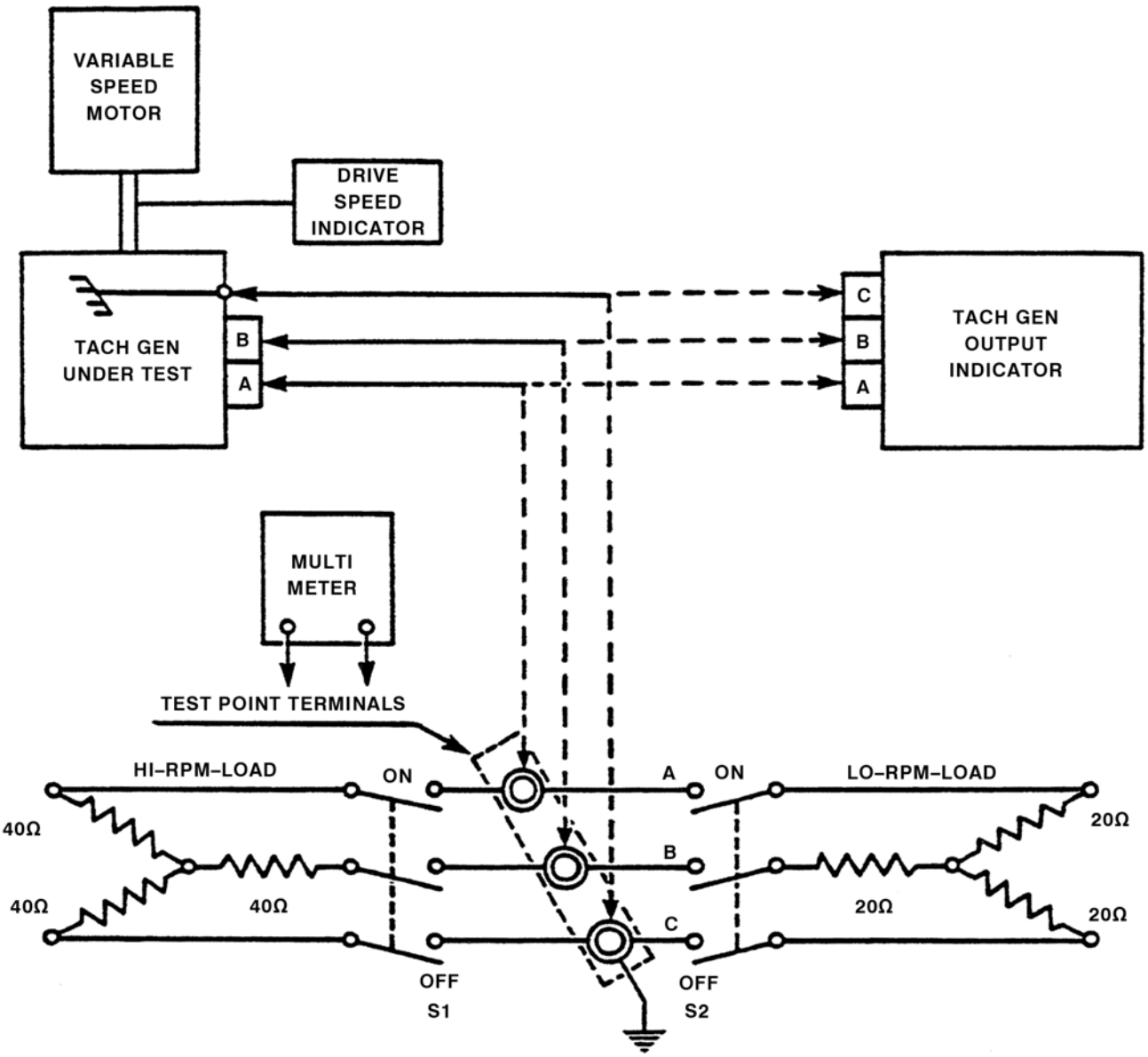
c) Operate motor and increase speed to 4200 rpm (drive speed indicator).

d) Set multimeter to 30 volts AC range and measure voltage outputs at the test point terminals across each of the phases A–B, A–C and B–C. The voltage across each phase shall be 21.0 ± 0.5 volts AC.

e) Decrease motor speed to 1000 rpm, select HI–RPM–LOAD switch to OFF and LO–RPM–LOAD switch to ON.

f) Repeat Para 1.A.3).d) and check phase voltages are not less than 3.5 volts AC.

g) Remove tachometer generator from test bench.



Test Rig Layout Schematic
Figure 1